WORKED SOLUTION

STEP I 2009, Question 5

Pure • Optimisation

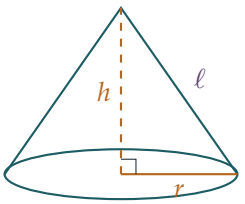
STEP I 2009, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q5

The key idea

There are three lengths but only two are free, since r, h and ℓ are tied together by Pythagoras: $\ell^2 = r^2 + h^2$. Holding one quantity fixed takes away one more freedom, so in each part we write the quantity to be made stationary as a function of r alone and differentiate. Square roots make the algebra messy, so it pays to work with V^2 or A^2 instead: these are stationary exactly where V or A is.



Solution

Throughout, r, h and ℓ are positive and, from the right-angled triangle in the cross-section,

$$\ell^2 = r^2 + h^2.$$

Since $V > 0$ and $A > 0$, the derivative of V^2 is $2V\dfrac{\mathrm{d}V}{\mathrm{d}r}$, which vanishes exactly when $\dfrac{\mathrm{d}V}{\mathrm{d}r}$ does; similarly for A. So we may find stationary points of the squares instead.

Part (i)

With A fixed, $\ell = \dfrac{A}{\pi r}$, so

$$h^2 = \ell^2 - r^2 = \frac{A^2}{\pi^2 r^2} - r^2.$$

(For a genuine cone we need $h > 0$, that is $A > \pi r^2$.) Then

$$V^2 = \tfrac{1}{9}\pi^2 r^4 h^2 = \tfrac{1}{9}\pi^2 r^4 \left(\frac{A^2}{\pi^2 r^2} - r^2\right)$$
$$= \tfrac{1}{9}\left(A^2 r^2 - \pi^2 r^6\right).$$

Differentiating with respect to r, keeping A constant,

$$\frac{\mathrm{d}(V^2)}{\mathrm{d}r} = \tfrac{1}{9}\left(2A^2 r - 6\pi^2 r^5\right) = \tfrac{2}{9} r\left(A^2 - 3\pi^2 r^4\right).$$

Since $r > 0$, this is zero exactly when

$$A^2 = 3\pi^2 r^4,$$

which is the first result. Taking the positive square root, $A = \sqrt{3}\,\pi r^2$, and so

$$\ell = \frac{A}{\pi r} = \frac{\sqrt{3}\,\pi r^2}{\pi r} = \sqrt{3}\, r,$$

as required. (Note that $A = \sqrt{3}\,\pi r^2 > \pi r^2$, so this really is a cone.)

The derivative of V^2 changes from positive to negative as r increases through this value, so the stationary value is a maximum: the cone of greatest volume for a given curved surface area. From $\ell^2 = r^2 + h^2$ we also get $h^2 = 3r^2 - r^2 = 2r^2$, so $h = \sqrt{2}\, r$.

Part (ii)

Now V is fixed, so $h = \dfrac{3V}{\pi r^2}$, and

$$A^2 = \pi^2 r^2 \ell^2 = \pi^2 r^2 (r^2 + h^2) = \pi^2 r^4 + \pi^2 r^2 \cdot \frac{9V^2}{\pi^2 r^4}$$
$$= \pi^2 r^4 + \frac{9V^2}{r^2}.$$

Differentiating with respect to r, keeping V constant,

$$\frac{\mathrm{d}(A^2)}{\mathrm{d}r} = 4\pi^2 r^3 - \frac{18V^2}{r^3},$$

which is zero exactly when

$$V^2 = \tfrac{2}{9}\pi^2 r^6, \qquad \text{so} \qquad V = \tfrac{\sqrt{2}}{3}\,\pi r^3$$

(taking the positive root, as $V > 0$).

Ask yourself We have found V in terms of r, but the question asks for h. How do we get there?

Use the volume formula once more: $V = \frac{1}{3}\pi r^2 h$. Setting this equal to $\frac{\sqrt{2}}{3}\pi r^3$ and cancelling $\frac{1}{3}\pi r^2$ gives

$$h = \sqrt{2}\,r.$$

This time the derivative goes from negative to positive (for small r the term $9V^2/r^2$ is huge, for large r the term $\pi^2 r^4$ is), so this is the cone of least curved surface area for a given volume.

? **Ask yourself** Is it a coincidence that both parts give the same shape, $h = \sqrt{2}\,r$?

No. If a cone maximises volume for its area, then it also minimises area for its volume: any cone with the same volume and smaller area could be scaled up to the original area, and it would then have a larger volume, which is impossible. Since the problem only fixes the shape up to scale, both parts must pick out the same ratio $h : r$.

Checks

- Take $r = 1$ in part (i): then $A = \sqrt{3}\,\pi$, $\ell = \sqrt{3}$, $h = \sqrt{2}$, and $\pi r \ell = \sqrt{3}\,\pi$ as it should be. Nudging r to 0.99 or 1.01 with A held at $\sqrt{3}\,\pi$ gives a slightly smaller volume.
- Numerically maximising V for random fixed A, and minimising A for random fixed V, always gave $h/r \approx 1.41421$, which is $\sqrt{2}$.
- Dimensions: A^2 and $\pi^2 r^4$ are both (length)^4; V^2 and $\pi^2 r^6$ are both (length)^6.

Common slips

- Forgetting that ℓ depends on r and h: treating ℓ as a constant in part (i), or h as a constant in part (ii), when in fact only A (or V) is fixed.
- Differentiating V directly with the square root still inside; it works, but it is easy to make an error. Squaring first avoids the chain rule entirely.

Beyond STEP

Finding the extreme value of one quantity while another is held fixed is a *constrained optimisation* problem. At university you meet the method of Lagrange multipliers, which handles the constraint without eliminating a variable, and which explains the "duality" between the two parts here. For a clear first treatment of multivariable calculus and Lagrange multipliers, try Stewart, *Calculus*; Siklos, *Advanced Problems in Mathematics*, has more STEP-style optimisation practice.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.